

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

```
1 #Q1, 1. Construct a 90% confidence interval for the mean IQ.
2 #step 1: For CI, must calculate the sample mean of IQs
3 y_samp_mean <- mean(y)
4 y_samp_mean #sample mean is 98.44
5
```

```

6 #step 2: then find the sd from the sample mean
7 y_sd <- sd(y)
8 y_sd #sd of y is 13.09287
9
10 #step 3: get n (sample size of students)
11 n <- length(y)
12 n #n = 25
13
14 #step 4: find standard error
15 y_se <- y_sd / sqrt(n)
16 y_se #standard error = 2.618575
17
18 #step 5: find the t-value to use in CI
19 alpha <- 0.10
20 t_val <- qt(1 - alpha/2, df = n-1)
21 t_val #t value = 1.710882
22
23 #step 6: use t-val to find margin of error with se
24 margin_error <- t_val * y_se
25 margin_error #margin_error = 4.480072
26
27 #step 7: use margin of error to find the lower and higher ends of CI
28 lower_end_CI <- y_samp_mean - margin_error
29 upper_end_CI <- y_samp_mean + margin_error
30 lower_end_CI #lower end CI is 93.95993
31 upper_end_CI #upper end CI is 102.9201
32 #CI is [93.96, 102.92]
33
34 #alternative route - I've discovered t.test
35 t.test(y, conf.level = 0.90)

```

The sample mean is 98.44, the sample standard deviation is 13.09, and the sample size is 25. Because the population standard deviation is unknown, I use the t-distribution with $df=n-1=24$.

For a 90% confidence interval, $\alpha=0.10$, so the t value is 1.71. The standard error is 2.619, and the margin of error is 4.48. Therefore, the confidence interval is 93.96 to 102.92.

So, I am 90% confident that the mean IQ of students at this school lies between 93.96 and 102.92.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```

1 #Q1, 2. Test if school's mean IQ is greater than 100 (alpha=0.05)
2 # One-sided t-test to find p val to test H(a) that school's mean > 100
3
4 #step 1: use the t statistic formula

```

```

5 t_stat <- (y_samp_mean - 100) / y_se
6 t_stat #t_stat = -0.5957439
7
8 #step 2: find the p-val without lower end (>100)
9 p_val <- pt(t_stat, df = n-1, lower.tail = FALSE)
10 p_val # p_val = 0.7215383
11
12 #answer: since p value 0.722 is greater than 0.05, fail to reject the
    null

```

The null hypothesis is that average student IQ in her school = 100, and alternative hypothesis is that it's greater than 100. This is a one-sided t-test. The test statistic is taken by comparing the sample mean to 100 and then dividing by standard error to get -0.596. Then using 24 degrees of freedom, the one-sided p-value is 0.722.

Since $p=0.722$ is greater than $\alpha=0.05$, I fail to reject the null hypothesis. There is insufficient evidence at the 5% significance level to conclude that the mean IQ of students at this school is greater than 100.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

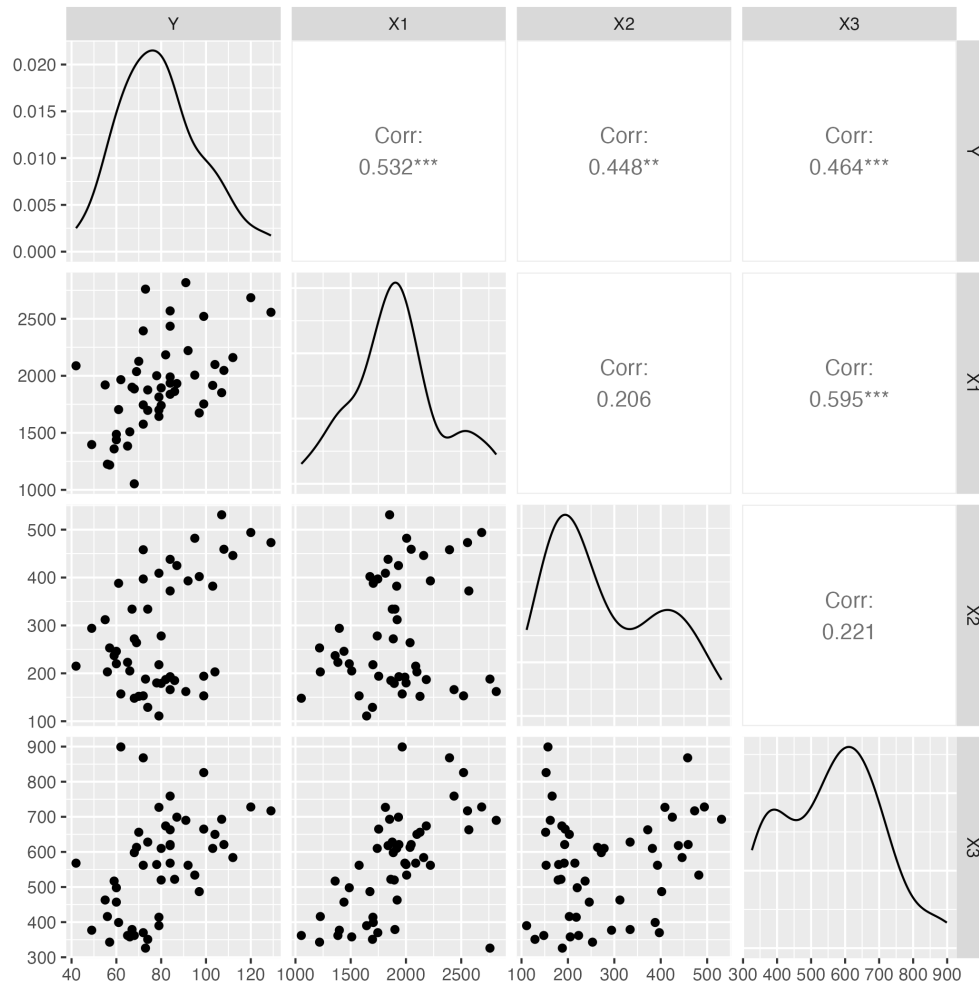
Explore the `expenditure` data set and import data into R.

```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

```
1 #explore data
2 View(expenditure)
3 summary(expenditure)
4 str(expenditure)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```
1 #Q2, 1. Plot relationships among vars. What are the correlations?
2 #step 1: subset the dataset to only look at the variables I want
3 graph_1 <- expenditure |>
4   select(Y, X1, X2, X3) |> #step 2: make scatterplot
5   ggpairs()
6
7 graph_1
```



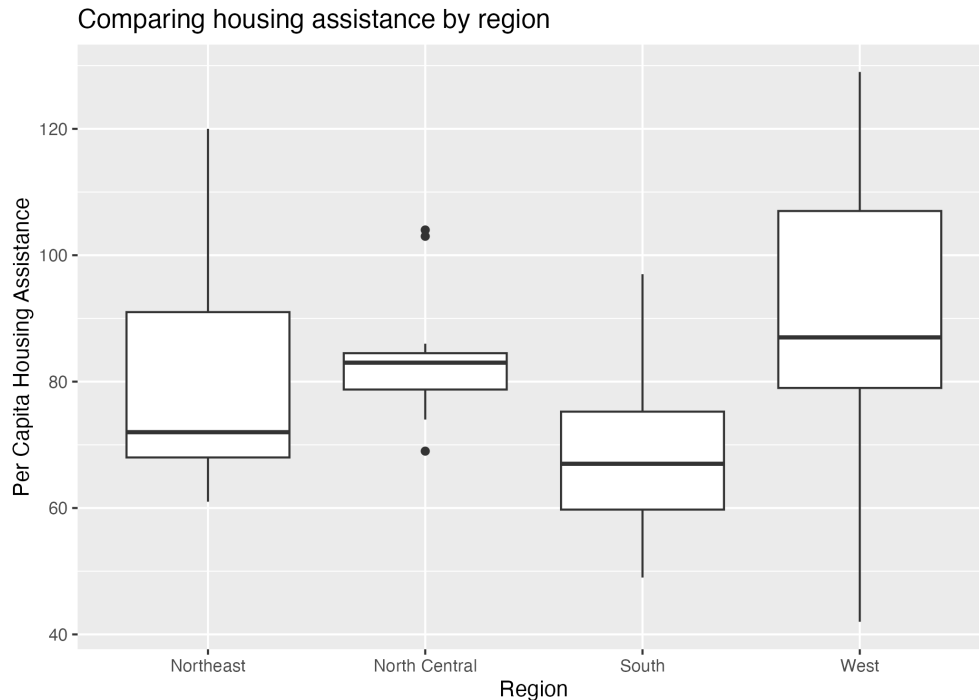
The scatterplots show positive relationships between all four variables. Y (per capita expenditure on housing assistance) has moderate positive relationships with $X1$ (per capita personal income) (0.532), $X2$ (the number of financially insecure residents per 100,000) (0.448), and $X3$ (the number of people per 1,000 in urban areas) (0.464). This means that higher values of each of these variables generally correspond with higher housing expenditures, although there is substantial variation. The strongest relationship is between $X1$ and $X3$, with a correlation of 0.595, suggesting that states with higher per capita personal income also tend to have a larger share of their population living in urban areas. The relationships between $X1$ and $X2$ (0.206) and between $X2$ and $X3$ (0.221) are comparatively weak. Overall, all of the relationships are positive, but none are especially strong, and the graphs show considerable spread around these general patterns.

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```

1 #Q2, 2. Plot relationship btwn Y and region. Which region has highest p/c
2   #expenditure on housing assistance
3
4 class(expenditure$Region)
5 #step 1: make region a factor
6 expenditure$Region <- factor(
7   expenditure$Region,
8   levels = c(1, 2, 3, 4),
9   labels = c("Northeast", "North Central", "South", "West")
10 )
11
12 #Step 1: y is numerical, region is categorical —> use boxplot
13
14 graph_y_region <- ggplot (expenditure, aes(x = Region, y = Y)) +
15   geom_boxplot() +
16   labs(
17     x = "Region",
18     y = "Per Capita Housing Assistance",
19     title = "Comparing housing assistance by region"
20   )
21
22 graph_y_region
23 #It appears that the West will have the highest, but let's check mean
24
25 #Step 2: calculate means by region
26 expenditure |>
27   group_by(Region) |>
28   summarise(mean_for_each = mean(Y))
29 #mean findings: NE = 79.4, NC = 83.9, S = 69.2, W = 88.3
30 # On average, the West has the highest average per capita expenditure on
31   housing
32 # assistance, with a mean of about 88.3.

```



The boxplot shows differences in housing assistance expenditure across the four regions, although there is also variation within each region. The South generally has lower expenditures, while the West and North Central regions tend to have higher expenditures.

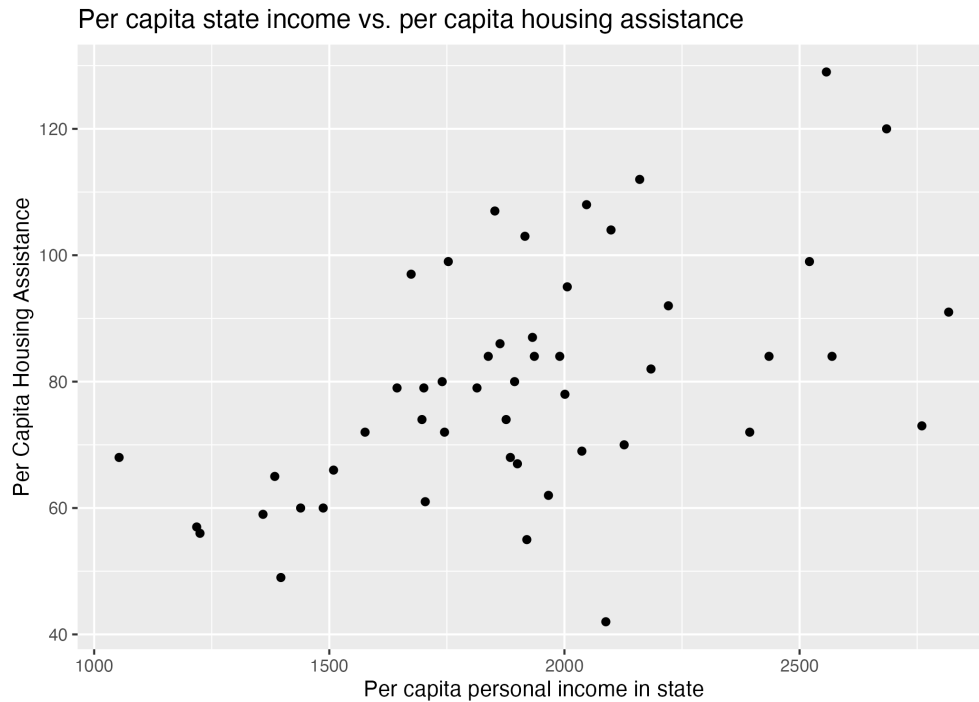
The mean per capita expenditure on housing assistance is approximately 79.4 in the Northeast, 83.9 in North Central, 69.2 in the South, and 88.3 in the West. Therefore, the West has the highest average per capita expenditure on housing assistance, with a mean of approximately 88.3.

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```

1 #Q2, 3. Plot relationship btwn Y and X1 Describe graph + relationship.
  Then,
2 #reproduce with Region and display w/ diff symbols and colors.
3 #Step 1: y is numerical, X1 is numerical —> use scatterplot
4 graph_y_X1 <- ggplot(expenditure, aes(x = X1, y = Y)) +
5   geom_point() +
6   labs(
7     title = "Per capita state income vs. per capita housing assistance",
8     x = "Per capita personal income in state",
9     y = "Per Capita Housing Assistance"
10  )
11
12 graph_y_X1

```



The scatterplot shows a moderate positive relationship between per capita personal income (X1) and per capita expenditure on housing assistance (Y). States with higher personal income generally tend to have higher housing assistance expenditures, although the points are fairly spread out, so the relationship is not especially strong.

```

1 #Step 2: add region with different symbols
2 graph_y_X1_region <- ggplot(expenditure,
3                               aes(x = X1, y = Y,
4                                   colour = Region, shape = Region)) +
5   geom_point() +
6   labs(
7     title = "Housing Assistance Expenditure and Personal Income by Region",
8     x = "Per capita personal income in state",
9     y = "Per Capita Housing Assistance"
10  )
11
12 graph_y_X1_region

```

